
THE POWER OF VACCINATION

From Smallpox Eradication to COVID-19 in Record Time

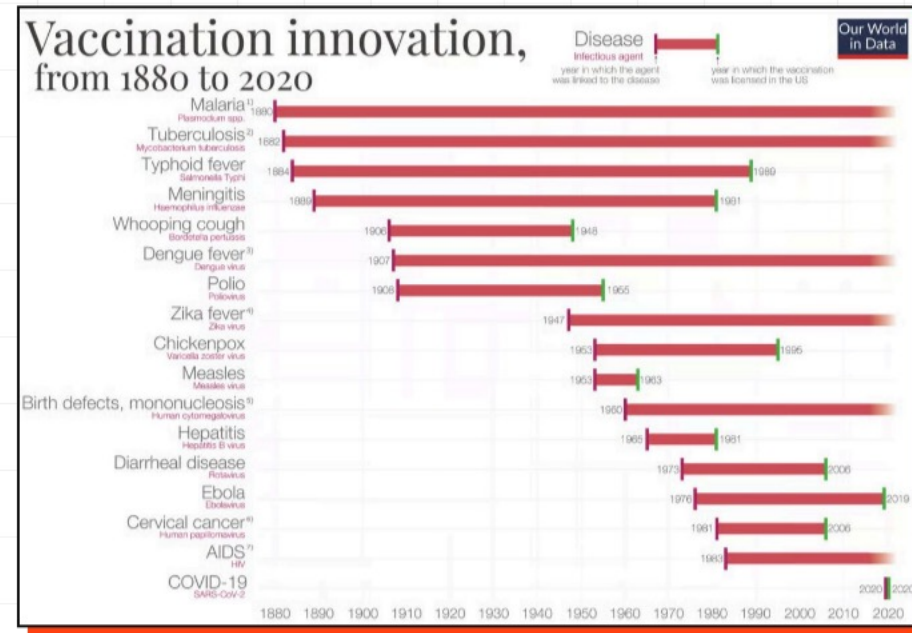
DATA SOURCE
Our World in Data (1796–2020)

From 100 Years to Under 1 Year

Vaccine Development Is Accelerating

The timeline from pathogen identification to vaccine licensure has collapsed, driven by mRNA technology and genomic tools.

Disease	Gap (Years)
Typhoid fever	105
Meningitis	92
Whooping cough	42
Hepatitis B	16
Measles	10
COVID-19	< 1



100% Case Reduction Achieved

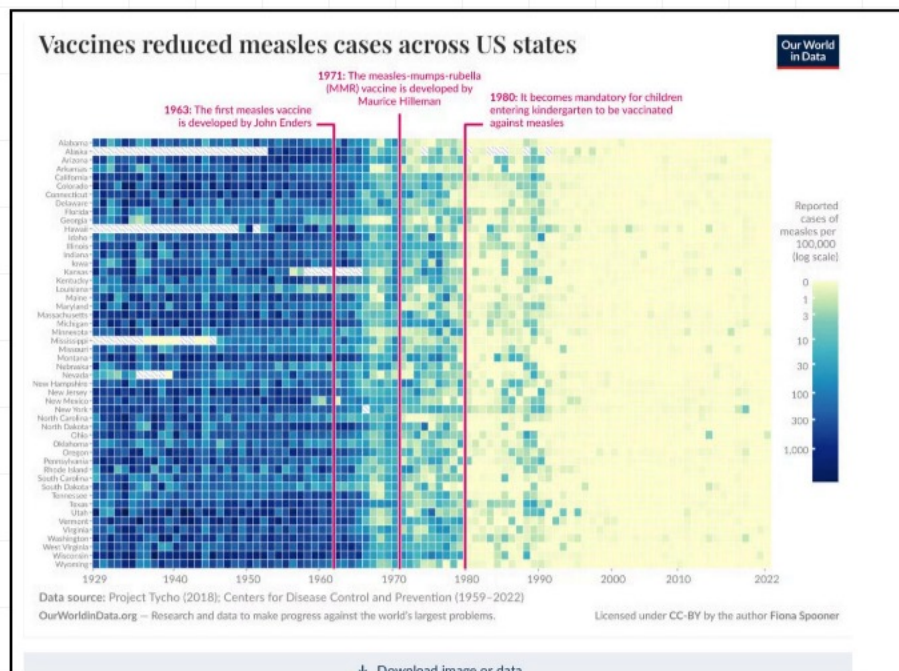
Diphtheria, Polio, and Smallpox in the US

Disease	Case Reduction	Death Reduction
Diphtheria	100%	100%
Smallpox	100%	100%
Acute polio	100%	100%
Measles	99.99%	100%
Rubella	99.98%	100%
Hepatitis A	89%	88.7%
Pneumococcal	40.5%	31.3%



Measles Cases Collapsed

Across All 50 US States After Vaccine Mandates



- Pre-1963: Near-universal high incidence (100–1,000+ cases per 100k)
- 1963: First measles vaccine developed
- 1971: MMR vaccine introduced
- 1980: Kindergarten mandate enforced

Result: Near zero cases nationwide post-1980.

Smallpox: The Only Human Disease Eradicated

Timeline

- 1959: WHO resolution to eradicate passed
- 1967: Intensified global campaign launched
- 1977: Last natural case (Somalia)
- 1980: Global eradication certified

Why It Worked

Virus infected only humans, symptoms were clearly distinguishable, and the vaccine was cheap and highly effective.

Herd Immunity

How Vaccinating Many Protects the Most Vulnerable

When a critical mass is vaccinated, pathogens cannot spread effectively.

This shields those who **cannot** be vaccinated:

- Newborn infants
- Immunocompromised individuals
- Cancer patients on chemotherapy



Gaps Persist

Impact is still incomplete for some diseases

Incomplete Control

- Pneumococcal disease: Only 40.5% case reduction, 31.3% death reduction
- Hepatitis A: 89% case reduction
- Acute Hepatitis B: 83.9% case reduction

Key Barriers

- Global access inequality
- Supply chain shortages
- Poor scientific understanding reducing trust
- Need for newer, more effective vaccines

Key Takeaways
